

Food Safety Intelligence System

User Guide & Service Overview

How it works · What you receive · Analytical methods · Industry landscape

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1. What is FSIS?

The Food Safety Intelligence System (FSIS) is a fully automated, AI-powered global pathogen recall monitoring platform built and operated by Advanced Food-Tech Solutions (AFTS). It tracks regulatory recall notices from **66 agencies across 60+ countries**, filters for pathogen-related hazards, and delivers structured intelligence to food safety professionals.

FSIS is not a raw data feed. Every recall is enriched, normalized, severity-ranked, and interpreted through the lens of food process engineering. The weekly and monthly briefings read like a report from a practicing process authority — naming specific failure modes, referencing regulatory frameworks (21 CFR 113/114, PMO, HACCP), and committing to concrete engineering recommendations.

You get intelligence, not data. The analysis speaks the language of food engineers, not the language of databases.

1.1 What's included

Feature	What you get
Live dashboard	Interactive intelligence platform with pathogen filtering, live feed, severity tiers, outbreak flags, cross-region comparisons. Always current.
Daily briefs	Auto-generated HTML page each morning showing yesterday's recalls worldwide, with Tier-1/outbreak triage and direct links to source regulatory notices.
Weekly digest	Every Friday: AI-written narrative with top 5 ranked threats, leading pathogen analysis, regional breakdown. Delivered to your inbox.
Monthly report	9-section analyst-grade briefing: trend analysis, hotspot matrix, outbreak timeline, concentration metrics, predictive models, engineering recommendations. With Appendix listing every recall. PDF attached.
Yearly report	Annual pathogen surveillance review: year-over-year trend analysis, geographic shifts, emerging-threat forecast for the year ahead. Shipped every December 31.

Custom alerts	Build rules on Pathogen × Product × Country × Brand (AND/OR logic). Get an email within minutes of a matching recall appearing. 2 rules included.
XLSX dataset	Full recall history — every pathogen, every country, every source. Download, query, chart, and build your own reports.

2. How it works

FSIS operates a continuous automated pipeline that runs 24/7 with no manual intervention. The pipeline has four stages: **collection, enrichment, analysis, and delivery**.

2.1 Collection

Automated scrapers connect to 66 regulatory agency feeds worldwide twice daily — a morning pass and an evening pass. Between these, multiple AI-powered gap-finders sweep for any recalls the scrapers might have missed, staggered across the day to catch postings from every time zone:

- 03:00 Athens — overnight sweep (APAC + Oceania catch-up)
- 12:00 Athens — midday worldwide sweep for missed recalls
- 14:00 Athens — EU afternoon catch-up · grounded regulator search
- 22:00 Athens — deterministic regulator-domain search (US end-of-day sweep — 15:00 ET / 12:00 PT)

2.2 Enrichment

Every raw recall is processed through an AI enrichment pipeline that normalizes pathogen names across languages (e.g. "Listeria monocytogenes", "Listeriose" all become "Listeria monocytogenes"), validates source URLs against the originating regulator's website, standardizes country and jurisdiction names, assigns severity tiers, and flags outbreak events.

A dedicated URL gate verifies that every recall links to a genuine regulatory source before it enters the published dataset. Recalls that cannot be traced to an official notice are held in a pending queue and re-checked automatically.

2.3 Analysis

- **Daily:** yesterday's recalls are compiled into a brief with Tier-1/outbreak triage and regional breakdown.
- **Weekly (Friday):** an AI-generated intelligence narrative covers the week's top threats, pathogen distribution, and regional patterns. The narrative is AI-generated and reviewed under the process authority of AFTS.
- **Monthly (1st):** a full 9-section report runs descriptive analytics (MoM trends, hotspot detection, cluster analysis, concentration metrics) and predictive models (trend projection, Poisson forecasts, change-point detection). The AI narrative interprets every finding through food process engineering principles.

2.4 Delivery

Reports are delivered automatically via email (HTML + PDF attachment for monthly) and are also available on the live dashboard. Custom alert matches trigger immediate email notification. All delivery uses deduplication (you never receive the same report twice), staleness checks (stale data is never sent), and one-click unsubscribe via a secure token link.

3. Daily briefs

A daily brief is generated every morning at 10:00 Athens time, covering the previous day's regulatory activity. It is a single HTML page showing:

- **KPI strip:** total recalls, Tier-1 critical count, outbreak count for the day.
- **Regional breakdown:** recalls grouped by region (Europe, North America, Latin America, Asia-Pacific, Middle East & Africa, Oceania).
- **Individual recall cards:** each showing the regulatory source, company name, product description, identified pathogen/hazard, and a direct link to the originating regulatory notice.
- **Tier classification:** every recall is labeled Tier 1 (immediate public health risk) or Tier 2 (significant but lower severity) based on the pathogen's clinical severity and the recall's outbreak status.

Daily briefs are accessible from the dashboard's **Daily** tab and are retained in a rolling 7-day archive.

4. Weekly intelligence digest

Every Friday, subscribers receive an email briefing covering the week's global pathogen recall activity. The digest includes:

- **KPI summary:** total recalls, Tier-1 count, outbreak count, and change vs prior week.
- **Leading pathogen card:** the dominant pathogen of the week with case count and percentage share.
- **AI intelligence summary:** a narrative paragraph interpreting the week's data through a food safety engineering lens — naming specific failure modes, production-system risks, and the single most important signal for QA attention.
- **Top 5 critical threats table:** severity-ranked incidents with pathogen, company, product, country, tier/outbreak badges, and direct links to regulatory notices.
- **Full HTML report link:** the complete week's report with all recalls, pathogen distribution, and geographic breakdown.

The weekly narrative is AI-generated under structured prompts that require process-authority-grade engineering language, and reviewed under the process authority of AFTS.

5. Monthly intelligence report

The monthly report is the flagship intelligence product — a 9-section analyst-grade briefing produced on the 1st of each month, covering the full prior calendar month. Subscribers receive it as an HTML email with a PDF attachment.

5.1 Intelligence narrative (§ 01)

Three paragraphs of AI-generated analysis, structured as:

1. **Month headline:** MoM direction, anomaly status, dominant pathogen share, most significant country×pathogen hotspot, composite severity score.
2. **Structural interpretation:** explains why the month looks the way it does. Identifies whether the signal is a single-country event, a multi-jurisdictional pattern, or an agency-concentrated data artifact. Ties the dominant pathogen to a specific failure mode (environmental harborage, cold-chain breach, thermal underprocess, post-process recontamination).

3. **Forward-looking recommendation:** names the single highest-leverage control a QA director should verify this month, grounded in the predictive model outputs.

When thermal-processing hazards appear in the month's data (e.g. *C. botulinum* in low-acid canned foods), a **Process Authority Note** is automatically appended, speaking in the specialized language of scheduled-process validation.

5.2 Descriptive analytics (§ 02–§ 06)

Month-over-month trend. The current month's recall count is compared against a rolling 3–6 month baseline using a Z-score. A $Z > 2$ flags the month as anomalously high; $Z < -2$ flags it as anomalously low. A sparkline chart visualizes the historical trajectory with color-coded direction.

Country × Pathogen hotspot matrix. A heatmap of observed recall counts by country and pathogen. Each cell is tested against the expected count under a statistical independence assumption. Cells where the observed count exceeds the expected by more than 2 standard deviations are flagged as hotspots — indicating a country-pathogen combination occurring more often than chance alone would predict.

Outbreak timeline & cluster analysis. A calendar-axis timeline showing outbreak events with pathogen-colored markers. A sliding 14-day window detects clusters: three or more same-pathogen outbreak events within any two-week span are flagged as a potential coordinated signal.

Regulatory concentration. Two metrics quantify signal structure. The Herfindahl-Hirschman Index (HHI) measures how concentrated the recall signal is among regulatory sources — a high HHI may indicate a single-agency activity spike rather than a genuine multi-jurisdictional event. The Gini coefficient measures geographic distribution inequality — helping distinguish between country-specific outbreaks and broad international patterns.

Emerging & declining pathogens. Each pathogen's current-month share is compared to its historical average. Pathogens whose share has grown by more than 2 standard deviations are flagged as emerging; those declining by more than 2 standard deviations are flagged as receding.

5.3 Predictive outlook (§ 07)

The report runs up to 8 predictive models. Models activate progressively as data history accumulates — an honest roadmap showing what is active today and what unlocks as the dataset grows. Currently active models:

Model	What it tells you
Linear trend (OLS)	Is the overall recall volume trending up, down, or flat? Next-month point forecast with 95% confidence interval.
Poisson forecast	For rare pathogens (e.g. <i>C. botulinum</i>): should we be worried about this month's count? Provides p90 and p95 upper bounds. If observed > p95, that's a genuine anomaly.
CUSUM detection	Has something structurally changed in the recall landscape, or is this just month-to-month noise? Detects sustained shifts that single-month comparisons would miss.

Models that will activate as data history grows (12–24 months required): seasonal decomposition (STL), OLS with seasonal dummies, Holt-Winters exponential smoothing, SARIMA, and Prophet. Each model's status, activation threshold, and output are displayed in the Predictive Outlook panel.

5.4 Top 10 & Appendix (§ 08 + A)

The Top 10 table ranks the month's most critical incidents by a multi-factor severity sort: outbreak status → *C. botulinum* priority → Tier-1 classification → pathogen severity score → date. The subscriber edition includes **Appendix A**: a complete register of every recall recorded in the month, in the same ranked format.

6. Yearly intelligence report

Generated on December 31st each year, the yearly report provides a long-range view of global pathogen recall activity:

- **Year-over-year trend analysis:** monthly recall volumes across the full available history, with annual comparisons and visual sparkline. As the dataset grows, multi-year patterns and long-range trajectories become visible.
- **Annual pathogen breakdown:** distribution shifts between years — which pathogens are growing, which are receding, which are new entrants.
- **Geographic shift detection:** countries and regions gaining or losing recall share year-over-year.
- **Regulatory agency activity:** which agencies increased or decreased publication frequency, and whether that reflects genuine food safety changes or agency-level operational factors.
- **Forward-looking forecast:** projected recall volumes, seasonal patterns, and emerging-threat candidates for the year ahead, based on all active predictive models.

7. Custom alerts

Subscribers can create personalized watch rules that trigger instant email notifications when a matching recall appears. Each rule combines up to two criteria from four dimensions:

- **Pathogen:** *Listeria*, *Salmonella*, *E. coli*/STEC, *C. botulinum*, and 20+ others.
- **Product category:** dairy, meat, seafood, fresh produce, infant formula, beverages, and 30+ categories following Codex Alimentarius classification.
- **Country/jurisdiction:** any of the 60+ monitored countries.
- **Brand/firm:** free-text matching against company and brand name fields.

Two criteria can be combined with AND logic ("*Listeria* in France") or OR logic ("*Listeria* or *Salmonella* anywhere"). The alert matcher runs daily against every new recall. Matches are emailed immediately with a formatted table and direct links to the regulatory notice. Each alert email includes a one-click manage/unsubscribe link.

8. Analytical methods summary

All statistics are computed deterministically from the validated Recalls dataset. The AI narrative receives these pre-computed signals as authoritative context and interprets them — it does not recompute or second-guess them.

Method	Technique	What it measures
MoM Z-score	(current – rolling_mean) / rolling_std	Is this month's volume normal or anomalous vs the recent baseline?

Hotspot residuals	Pearson chi-square standardized residuals on the country×pathogen contingency table	Which country-pathogen combinations are over-represented vs independence?
Cluster detection	14-day sliding window, ≥3 same-pathogen events	Are outbreak events sporadic or temporally linked?
Source HHI	Herfindahl-Hirschman Index on source shares	Is the signal coming from many agencies or dominated by one?
Geographic Gini	Gini coefficient on country recall distribution	Are recalls spread globally or concentrated in one region?
Emerging pathogens	Z-score on current-month share vs historical mean share	Which pathogens are growing faster than historical norms?
Severity index	Weighted composite (volume, Tier-1 share, outbreaks, hotspots)	Single 0–100 score summarizing the month's overall threat level.
Linear trend	OLS regression on monthly counts	Next-month forecast with 95% CI and slope direction.
Poisson forecast	Poisson CDF on per-pathogen monthly counts	Upper bounds (p90, p95) for rare pathogens. "Is 3 <i>C. botulinum</i> recalls abnormal?"
CUSUM	Page (1954) tabular cumulative sum	Has the mean recall rate shifted structurally?

9. FSIS in the food safety intelligence landscape

FSIS operates alongside several established platforms that serve the broader food safety and regulatory intelligence space. Each platform has distinct strengths that serve different needs. We see them as fellow contributors to a shared goal: making the global food supply safer through better information.

Platform	Focus	Strengths	FSIS complements by...
RASFF (EU)	EU/EEA regulatory alerts	The authoritative source for EU food and feed safety alerts. Comprehensive, official, trusted by regulators worldwide.	Aggregating RASFF with 60+ non-EU sources into one global view. Many FSIS users monitor RASFF directly — FSIS adds the rest of the world.
HorizonScan (Fera Science)	Global hazard monitoring	Broad hazard coverage beyond pathogens (pesticides, contaminants, fraud). Backed by UK government science. Excellent for horizon scanning across all food risks.	Offering deeper pathogen-specific analysis: AI narrative with engineering interpretation, predictive models, and process-authority voice.

FoodAKAI	EU food fraud + safety prediction	Pioneering ML-based prediction of food fraud and safety events using RASFF data. Strong academic foundation and innovative risk-scoring algorithms.	Extending geographic scope beyond the EU, and focusing on pathogen-specific engineering interpretation rather than fraud prediction.
SGS Digicomply	Enterprise regulatory intelligence	Full-scale compliance platform with regulatory change tracking, supplier management, audit workflows. Designed for enterprise food safety teams with dedicated compliance staff.	Serving the market SGS does not target: individual QA managers, small-to-mid food companies, consultants, and process authorities who need intelligence without enterprise infrastructure.
FDA iRES / USDA FSIS	US regulatory feeds	The definitive source for US recalls. OpenFDA API enables programmatic access. USDA FSIS provides RSS and email alerts for meat/poultry/egg products.	Adding 60+ non-US sources alongside FDA/USDA data, plus AI narrative that interprets US recalls in a global context. FSIS ingests both FDA and USDA as primary sources.
INFOSAN (WHO)	International outbreak coordination	The gold standard for international food safety event coordination between national authorities. Confidential inter-governmental communication channel.	Operating in the public-data layer. INFOSAN coordinates between governments; FSIS serves the private sector by monitoring what governments publish. Complementary roles.

We believe more monitoring makes everyone safer. FSIS adds a pathogen-focused, AI-interpreted, process-authority-voiced layer to a landscape where each platform has earned its place.

10. Data sources

Regulatory agencies

FDA (US), USDA FSIS (US), EU RASFF, FSA (UK), FSANZ (Australia/New Zealand), CFIA (Canada), RappelConso (France), BVL (Germany), AESAN (Spain), EFET (Greece), MPI (New Zealand), SFA (Singapore), CFS (Hong Kong), and 50+ additional national authorities across Europe, Asia-Pacific, Latin America, the Middle East, and Africa.

Pathogen scope

Salmonella, Listeria, E. coli/STEC, C. botulinum, Norovirus, Hepatitis A, Campylobacter, Vibrio, Cyclospora, Yersinia, Bacillus cereus, Brucella, Cronobacter, Staphylococcus aureus, aflatoxin, ochratoxin A, patulin, histamine/scombrototoxin, marine biotoxins, mycotoxins, physical hazards (glass, metal, plastic fragments), chemical contaminants (rodenticides, heavy metals), and pest contamination.

Exclusions

Allergen-only recalls, labeling errors, quality issues (taste, texture, appearance), drug contamination, and product tampering are excluded from the FSIS scope. These are important food safety categories but fall outside the pathogen surveillance mission.

Process authority

Analytical frameworks, severity rubrics, pathogen classification, and the engineering interpretation of each recall are developed under the process authority of AFTS, drawing on in-house expertise in food process engineering, thermal processing, and regulatory compliance. Every view is grounded in validated process engineering: thermal processing (21 CFR 113/114), pasteurization (PMO), aseptic and UHT, hold-tube and F-value lethality, and HACCP.

Advanced Food-Tech Solutions · AFTS

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